

Financial Statement Analysis

Session 1 — Working with Financial Statements

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- Identify the different types of financial statements and their main components.
- Explain the stakeholders who use financial statements and their respective interests.
- Calculate and interpret key financial ratios to assess a company's condition and performance.
- Apply common-size analysis and benchmarking (time-trend & peer group) as analytical tools.

- 1 Types of Financial Statements
- 2 Users of Financial Statements
- 3 Financial Ratios
- 4 Summary

What Are Financial Statements?

Financial statements are a systematic summary of a company's financial transactions, prepared according to accounting standards (in Indonesia: PSAK/IFRS) for a given period.

There are four core statements essential to financial analysis:

- ① Balance Sheet (Statement of Financial Position)
- ② Income Statement
- ③ Statement of Cash Flows
- ④ Statement of Changes in Equity

1. Balance Sheet (Statement of Financial Position)

The balance sheet shows a company's financial position at **one point in time** (a snapshot).

Basic Accounting Identity

$$\text{Assets} = \text{Liabilities} + \text{Shareholders' Equity}$$

Key concept

Assets are listed by liquidity (most → least liquid); liabilities and equity by maturity order.

Net Working Capital = Current Assets – Current Liabilities — a key short-term liquidity indicator.

Balance Sheet — Main Components

Component	Description	Example Accounts
Current Assets	Expected to convert to cash within ≤ 1 year / one operating cycle	Cash, receivables, inventory, prepaid expenses
Fixed / Non-current Assets	Tangible or intangible, economic life > 1 year	Land, buildings, machinery, right-of-use assets, goodwill
Current Liabilities	Due within ≤ 1 year	Accounts payable, short-term bank debt, accrued expenses
Long-term Debt	Due in > 1 year	Bonds, long-term bank loans, long-term lease liabilities
Shareholders' Equity	Residual claim on assets after liabilities	Share capital, paid-in capital, retained earnings

2. Income Statement

The income statement measures performance **over a period of time** (a flow), unlike the balance sheet's snapshot.

Top-Down Structure

$$\text{Revenue (Sales)} - \text{COGS} = \text{Gross Profit}$$
$$\text{Gross Profit} - \text{Operating Expenses} = \text{EBIT}$$
$$\text{EBIT} - \text{Interest Expense} = \text{EBT}$$
$$\text{EBT} - \text{Taxes} = \text{Net Income}$$

- **Non-cash items:** depreciation & amortization reduce accounting profit but are not a cash outflow.
- **Interest expense** is a financing cost, kept separate from operating costs so pure operating performance can be assessed.
- **Earnings per Share (EPS)** = $\text{Net Income} \div \text{Shares Outstanding}$.
- **Dividends per Share (DPS)** and the **dividend payout ratio** are commonly reported as supplementary information.

3. Statement of Cash Flows

Reconciles accrual-basis accounting profit with actual cash received/paid, explaining why the cash balance changed between periods.

Category	Content & Example Transactions
Operating Activities	Cash from core business: receipts from customers, payments to suppliers & employees, interest and tax payments
Investing Activities	Purchase/sale of fixed assets (capex), business acquisitions, purchase/sale of long-term investments
Financing Activities	Issuance/repayment of debt, issuance/buyback of shares, dividend payments

Cash Flow from Assets (CFFA)

Ross's framework linking the operating and financing sides of the firm.

Formula

$$\text{CFFA} = \text{Operating Cash Flow} - \text{Net Capital Spending} - \Delta \text{Net Working Capital}$$

$$\text{CFFA} = \text{Cash Flow to Creditors} + \text{Cash Flow to Stockholders}$$

Why it matters

High net income does not guarantee cash availability — a company can be profitable on paper yet face a liquidity crisis (“profit $\neq$ cash”). The cash flow statement is an early-warning tool for solvency problems.

4. Equity Statement and Notes

Statement of Changes in Equity

- Details movements in each equity component: opening balance, additions (net income, new capital), deductions (dividends, buybacks), closing balance.

Notes to the Financial Statements

- Explain accounting policies, assumptions, estimates, account details, contingencies, related-party transactions, and subsequent events.
- Analysts must read the notes — much material information (e.g., off-balance-sheet commitments, litigation) is disclosed only here.

- **Historical Cost Principle** — assets are generally recorded at acquisition cost, not current market value.
- **Accrual / Matching Principle** — revenue is recognized when earned, and expenses are matched to related revenue in the same period.
- **Going Concern Assumption** — statements assume the entity will continue operating in the future.

Implication for analysts

Because statements are accrual-based and historical-cost, **book value** often differs from **market value** — market value is more relevant for investment decisions since it reflects expected future cash flows.

Who Uses Financial Statements?

Financial statements are general purpose — prepared to meet the information needs of many stakeholders with different interests.

Broadly split into **internal** and **external** users.

Party	Main Interest
Management (Directors & Managers)	Planning, control, and operational/strategic decisions; basis for evaluating divisional performance
Employees & Labor Unions	Assessing business stability and going-concern (job security); ability to pay wages, bonuses, pensions
Board of Commissioners / Audit Committee	Oversight of management performance and compliance with corporate governance (GCG)

Party	Main Interest
Investors / Shareholders	Assessing profitability, risk, and return prospects; basis for buy/hold/sell decisions
Short-term Creditors (Trade/Suppliers)	Assessing liquidity — ability to pay short-term obligations on time
Long-term Creditors (Banks, Bondholders)	Assessing long-term solvency and capital structure; covenant compliance
Tax Authority (e.g., DJP)	Verifying tax obligations and compliance

External Users (2/2)

Party	Main Interest
Capital Market Regulators (OJK, IDX) Customers	Ensuring disclosure compliance and protecting public investors Assessing supplier continuity for long-term contracts or supply dependency
Financial Analysts & Rating Agencies	Investment recommendations and credit ratings based on ratios and projections
External Auditors	Independent opinion on the fairness of the financial statements
Public & Competitors	Economic impact assessment and competitive benchmarking

Class discussion

Different stakeholders emphasize different parts of the financial statements: short-term creditors focus on the balance sheet (liquidity); shareholders focus on earnings & free cash flow (profitability and valuation); long-term lenders focus on capital structure and cash flow coverage.

Why Use Financial Ratios?

Financial ratios turn raw financial-statement figures into comparable measures — across time (**time-trend analysis**) or across peer companies (**peer group analysis**) — by neutralizing differences in company size.

Ross groups ratios into **five** main categories:

- ① Liquidity (short-term solvency)
- ② Long-term solvency (leverage)
- ③ Asset management (turnover / activity)
- ④ Profitability
- ⑤ Market value

Liquidity Ratios

Measure a company's ability to meet short-term obligations without financial strain.

Ratio	Formula	Interpretation
Current Ratio	$\text{Current Assets} \div \text{Current Liabilities}$	Higher $\rightarrow$ better liquidity, but too high may mean idle cash
Quick Ratio (Acid-Test)	$(\text{Current Assets} - \text{Inventory}) \div \text{Current Liabilities}$	More conservative; excludes the least-liquid asset
Cash Ratio	$\text{Cash \& Equivalents} \div \text{Current Liabilities}$	Strictest liquidity measure

Long-Term Solvency (Leverage) Ratios

Measure the extent of debt usage and ability to meet long-term obligations.

Ratio	Formula	Interpretation
Total Debt Ratio	$(\text{Total Assets} - \text{Total Equity}) \div \text{Total Assets}$	Proportion of assets financed by debt
Debt-to-Equity Ratio	$\text{Total Debt} \div \text{Total Equity}$	Higher $\rightarrow$ more aggressive capital structure
Equity Multiplier	$\text{Total Assets} \div \text{Total Equity}$	$= 1 + \text{Debt-Equity Ratio}$
Times Interest Earned	$\text{EBIT} \div \text{Interest Expense}$	Ability of operating earnings to cover interest
Cash Coverage Ratio	$(\text{EBIT} + \text{D\&A}) \div \text{Interest Expense}$	More accurate than TIE; adds back non-cash items

Asset Management (Turnover) Ratios

Measure how efficiently a company manages and utilizes its assets to generate sales.

Ratio	Formula	Interpretation
Inventory Turnover	$\text{COGS} \div \text{Inventory}$	How many times inventory is sold & replaced per year
Days' Sales in Inventory	$365 \div \text{Inventory Turnover}$	Average days inventory sits before sale
Receivables Turnover	$\text{Sales} \div \text{Accounts Receivable}$	Speed of collecting receivables
Days' Sales in Receivables (DSO)	$365 \div \text{Receivables Turnover}$	Average collection period
Total Asset Turnover	$\text{Sales} \div \text{Total Assets}$	Efficiency of using total assets to generate sales

Profitability Ratios

Ratio	Formula	Interpretation
Profit Margin	$\text{Net Income} \div \text{Sales}$	% of sales converted to net income
Return on Assets (ROA)	$\text{Net Income} \div \text{Total Assets}$	Efficiency at generating profit from assets
Return on Equity (ROE)	$\text{Net Income} \div \text{Total Equity}$	Return to shareholders — the metric investors watch most

Dupont Identity — Decomposing ROE

ROE can be decomposed into three drivers to diagnose the source of performance.

Dupont Formula

$$\text{ROE} = \text{Profit Margin} \times \text{Total Asset Turnover} \times \text{Equity Multiplier}$$

$$\text{ROE} = \frac{\text{Net Income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Total Assets}} \times \frac{\text{Total Assets}}{\text{Total Equity}}$$

- Profit margin → operating efficiency & cost control
- Total asset turnover → efficiency of asset use
- Equity multiplier → capital structure (leverage) policy

Insight

Two companies can have the same ROE yet be driven by very different factors — e.g., thin margin with high turnover (retail) vs. thick margin with low turnover (real estate). High leverage can boost ROE but also raises financial risk.

Link accounting data with stock market prices — relevant only for publicly listed

companies.	Ratio	Formula	Interpretation
	Price–Earnings Ratio (P/E)	Price per Share $\div$ EPS	Growth-expectation proxy
	Market-to-Book Ratio	Market Price $\div$ Book Value per Share	>1: market values the firm above accounting value
	EV/EBITDA	Enterprise Value $\div$ EBITDA	Capital-structure-neutral valuation metric

Using Ratios: Benchmarking & Common-Size Statements

- ① **Time-Trend Analysis** — compare a company's own ratios over time.
- ② **Peer Group Analysis** — compare ratios with similar companies in the same industry.
- ③ **Common-Size Statements** — express each line as a % of total assets (balance sheet) or sales (income statement).

Methodological caution

Ratios are descriptive, not prescriptive — there is no universal “ideal number”; reasonable benchmarks differ by industry.

- Four core statements: Balance Sheet (position snapshot), Income Statement (period performance), Statement of Cash Flows (actual cash movement), Statement of Changes in Equity.
- Different users of financial statements focus on different aspects, based on their respective interests.
- Five ratio categories complement each other to form a complete picture; the Dupont Identity bridges profitability, efficiency, and leverage.

- ① Why can a company report positive net income yet still face a cash crunch?
- ② As a bank credit analyst evaluating a long-term loan, which ratio category would you prioritize, and why?
- ③ Two companies in the same industry have identical ROE. How does the Dupont Identity help determine which is fundamentally healthier?
- ④ What is the risk of relying on a single ratio without considering industry context and time trends?

Ross, S. A., Westerfield, R. W., Jaffe, J., & Jordan, B. D. *Corporate Finance* — Chapter 3: Working with Financial Statements.

For the Indonesian context, compare with the financial statement presentation format under **PSAK 1** (Presentation of Financial Statements) and **PSAK 2** (Statement of Cash Flows).

Thank You

Questions & Discussion